LAB REPORT: 04

DCCN

SUBMITTED TO

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BSC-IET 3RD

LAB REPORT: 04

Static Routing on Cisco Packet Tracer

Lab Task:

Make 4 router topology with 4 Pc's and ping it and do network advertised missing Ip's.

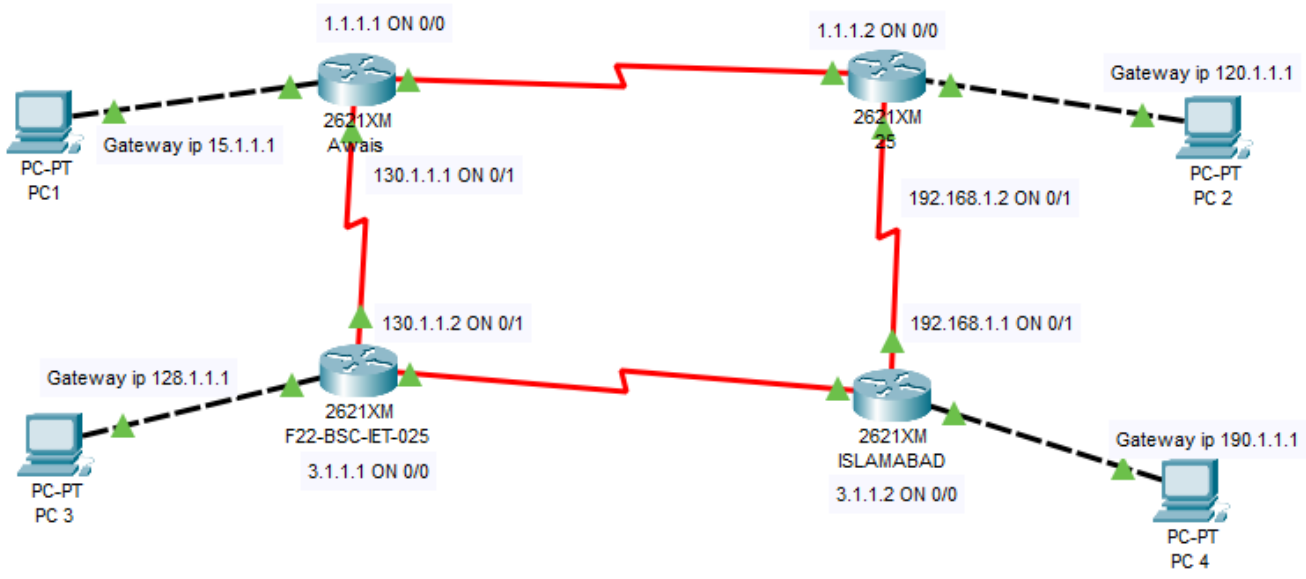
OBJECTIVE

The objective of this lab experiment was to design and implement a complex network topology comprising four routers and multiple PCs. The primary goals included configuring IP addresses, establishing network connectivity between routers and PCs, implementing routing protocols, and ensuring seamless communication within the network.

RESOURCES REQUIRED:

- 4 Routers (2621)
- 4 Pc's
- Serial 0/0, 0/1
- Fast Ethernet 0/0

TOPOLOGY OF 4 ROUTER AND 4 PC'S



Method

First switch off all router and put wic-2T card then switch on and configure Ip's to router

Router 1 Ip on 0/0.

```
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Awais
Awais(config)#int serial 0/0
Awais(config-if)#ip add 1.1.1.1 255.0.0.0
Awais(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0, changed state to down
```

Router 2 Ip on 0/0

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname 25
25(config)#int serial 0/0
25(config-if)#ip add 1.1.1.2 255.0.0.0
25(config-if)#no shutdown
```

Now serial 0/0 is green on router 1 and router 2 it means its connected

Router 1 Ip on 0/1

```
Awais(config-if)#int serial 0/1
Awais(config-if)#ip address 130.1.1.1 255.255.0.0
Awais(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1, changed state to down
Awais(config-if)#
```

Router 3 Ip on 0/1

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname F22-BSC-IET-025
F22-BSC-IET-025(config)#INT SERIAL 0/1
F22-BSC-IET-025(config-if)#ip add 130.1.1.2 255.255.0.0
F22-BSC-IET-025(config-if)#no shutdown

F22-BSC-IET-025(config-if)#
%LINK-5-CHANGED: Interface Serial0/1, changed state to up
```

Now serial 0/1 is green on router 1 and router 3 it means its connected

Router 2 Ip on 0/1

```
25(config-if)#int serial 0/1
25(config-if)#ip add 192.168.1.1 255.255.255.0
25(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1, changed state to down
25(config-if)#
```

Router 4 Ip on 0/1

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname ISLAMABAD
ISLAMABAD(config)#int serial 0/1
ISLAMABAD(config-if)#ip add 192.168.1.2 255.255.255.0
ISLAMABAD(config-if)#no shutdown

ISLAMABAD(config-if)#
%LINK-5-CHANGED: Interface Serial0/1, changed state to up
```

Now serial 0/1 is green on router 2 and router 4 it means its connected

Router 3 Ip on 0/0

```
F22-BSC-IET-025#
F22-BSC-IET-025#config t
Enter configuration commands, one per line.  End with CNTL/Z.
F22-BSC-IET-025(config)#int serial 0/0
F22-BSC-IET-025(config-if)#ip add 3.1.1.1 255.0.0.0
F22-BSC-IET-025(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0, changed state to down
F22-BSC-IET-025(config-if)#
```

Router 4 Ip on 0/0

```
ISLAMABAD#config t
Enter configuration commands, one per line.  End with CNTL/Z.
ISLAMABAD(config)#int serial 0/0
ISLAMABAD(config-if)#ip add 3.1.1.2 255.0.0.0
ISLAMABAD(config-if)#^Z
```

Now serial 0/1 is green on router 2 and router 4 it means its connected

Advertised missing Ip's in router 1:

R3 0/0 MISSING IN R1

```
Awais#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Awais(config)#ip route 3.0.0.0 255.0.0.0 130.1.1.2
Awais(config)#^Z
Awais#
%SYS-5-CONFIG_I: Configured from console by console
```

R2 0/1 MISSING IN R1

```
Awais#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Awais(config)#ip route 192.168.1.0 255.255.255.0 1.1.1.2
Awais(config)#^Z
Awais#
```

R4 0/0 AND 0/1 MISSING IN R1

```
Awais#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Awais(config)#ip route 3.0.0.0 255.0.0.0 3.1.1.1
Awais(config)#ip route 192.168.1.0 255.255.255.0 3.1.1.1
Awais(config)#^Z
```

SHOW IP ROUTE ON ROUTER 1:

```
Awais#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA
       N1 - OSPF NSSA external type 1, N2 - OSPF NS
       E1 - OSPF external type 1, E2 - OSPF externa
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS le
       * - candidate default, U - per-user static r
       P - periodic downloaded static route

Gateway of last resort is not set

C    1.0.0.0/8 is directly connected, Serial0/0
S    3.0.0.0/8 [1/0] via 130.1.1.2
               [1/0] via 1.1.1.2
C    130.1.0.0/16 is directly connected, Serial0/1
S    192.168.1.0/24 [1/0] via 1.1.1.2
                    [1/0] via 3.1.1.1
```

Advertised missing Ip's in router 2:

R1 0/1 MISSING IN R2

```
25#config t
Enter configuration commands, one per line.  End with CNTL/Z.
25(config)#ip route 130.1.0.0 255.255.0.0 1.1.1.1
25(config)#^Z
25#
```

R4 0/0 MISSING IN R2

```
25#config t
Enter configuration commands, one per line.  End with CNTL/Z.
25(config)#ip route 3.0.0.0 255.0.0.0 192.168.1.2
25(config)#^Z
25#
```

R3 0/0 AND 0/1 MISSING IN R2

```
25#config t
Enter configuration commands, one per line.  End with CNTL/Z.
25(config)#ip route 3.0.0.0 255.0.0.0 3.1.1.2
25(config)#ip route 130.1.0.0 255.255.0.0 3.1.1.2
25(config)#^Z
25#
```

SHOW IP ROUTE ON ROUTER 2:

```
25>
25>EN
25#SHOW IP ROUTE
Codes: C - connected, S - static, I - IGRP, R - RIP, M
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - C
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA e
       E1 - OSPF external type 1, E2 - OSPF external ty
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-
       * - candidate default, U - per-user static route
       P - periodic downloaded static route

Gateway of last resort is not set

C    1.0.0.0/8 is directly connected, Serial0/0
S    3.0.0.0/8 [1/0] via 192.168.1.2
S    130.1.0.0/16 [1/0] via 1.1.1.1
                  [1/0] via 3.1.1.2
C    192.168.1.0/24 is directly connected, Serial0/1
```

Advertised missing Ip's in router 3:

R1 0/0 MISSING IN R3

```
F22-BSC-IET-025#config t
Enter configuration commands, one per line.  End with CNTL/Z.
F22-BSC-IET-025(config)#ip route 1.0.0.0 255.0.0.0 130.1.1.1
F22-BSC-IET-025(config)#^Z
F22-BSC-IET-025#
```

R4 0/1 MISSING IN R3

```
F22-BSC-IET-025(config)#ip route 192.168.1.0 255.255.255.0 3.1.1.2
F22-BSC-IET-025(config)#^Z
F22-BSC-IET-025#
%SYS-5-CONFIG_I: Configured from console by console
```

R2 0/0 AND 0/1 MISSING IN R3

```
F22-BSC-IET-025#config t
Enter configuration commands, one per line.  End with CNTL/Z.
F22-BSC-IET-025(config)#ip route 1.0.0.0 255.0.0.0 1.1.1.1
F22-BSC-IET-025(config)#ip route 192.168.1.0 255.255.255.0 1.1.1.1
F22-BSC-IET-025(config)#^Z
F22-BSC-IET-025#
```

SHOW IP ROUTE ON ROUTER 3:

```
F22-BSC-IET-025#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA
       N1 - OSPF NSSA external type 1, N2 - OSPF NS
       E1 - OSPF external type 1, E2 - OSPF externa
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS le
       * - candidate default, U - per-user static r
       P - periodic downloaded static route

Gateway of last resort is not set

S    1.0.0.0/8 [1/0] via 130.1.1.1
C    3.0.0.0/8 is directly connected, Serial0/0
C    130.1.0.0/16 is directly connected, Serial0/1
S    192.168.1.0/24 [1/0] via 3.1.1.2
                    [1/0] via 1.1.1.1
```

Advertised missing Ip's in router 4:

R3 0/1 MISSING IN R4

```
ISLAMABAD#config t
Enter configuration commands, one per line.  End with CNTL/Z.
ISLAMABAD(config)#ip route 130.1.0.0 255.255.0.0 3.1.1.1
ISLAMABAD(config)#^Z
```

R2 0/0 MISSING IN R4

```
ISLAMABAD#config t
Enter configuration commands, one per line.  End with CNTL/Z.
ISLAMABAD(config)#ip route 1.0.0.0 255.0.0.0 1.1.1.2
ISLAMABAD(config)#^Z
ISLAMABAD#
```

R1 0/0 AND 0/1 MISSING IN R4

```
ISLAMABAD>en
ISLAMABAD#config t
Enter configuration commands, one per line.  End with CNTL/Z.
ISLAMABAD(config)#ip route 1.0.0.0 255.0.0.0 3.1.1.1
ISLAMABAD(config)#ip route 130.1.0.0 255.255.0.0 3.1.1.1
ISLAMABAD(config)#^Z
ISLAMABAD#
```

SHOW IP ROUTE ON ROUTER 4:

```
ISLAMABAD#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M
       D - EIGRP, EX - EIGRP external, O - OSPF, IA -
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA
       E1 - OSPF external type 1, E2 - OSPF external t
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level
       * - candidate default, U - per-user static rout
       P - periodic downloaded static route

Gateway of last resort is not set

S    1.0.0.0/8 [1/0] via 192.168.1.1
               [1/0] via 3.1.1.1
C    3.0.0.0/8 is directly connected, Serial0/0
S    130.1.0.0/16 [1/0] via 3.1.1.1
C    192.168.1.0/24 is directly connected, Serial0/1
```

Configure pc Ip in router fast Ethernet 0/0

Pc1 with R1

```
Awais#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Awais(config)#int fast
Awais(config)#int fastEthernet 0/0
Awais(config-if)#ip add 2.1.1.2 255.0.0.0
Awais(config-if)#no shutdown
```

Pc2 with R2

```
25(config)#config t
%Invalid hex value
25(config)#int fast
25(config)#int fastEthernet 0/0
25(config-if)#ip add 120.1.1.1 255.255.0.0
25(config-if)#no shutdown
```

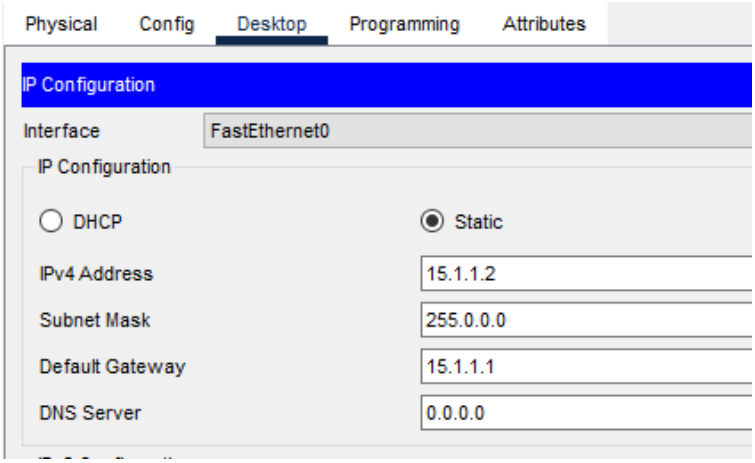
Pc3 with R3

```
F22-BSC-IET-025#config t
Enter configuration commands, one per line.  End with CNTL/Z.
F22-BSC-IET-025(config)#int fast
F22-BSC-IET-025(config)#int fastEthernet 0/0
F22-BSC-IET-025(config-if)#ip add 128.1.1.1 255.255.0.0
F22-BSC-IET-025(config-if)#no shutdown
```

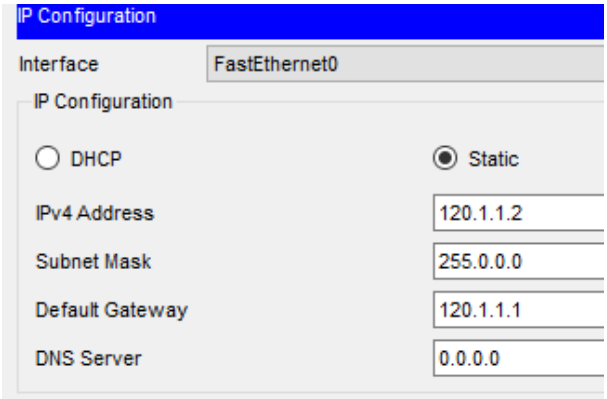
Pc4 with R4

```
ISLAMABAD(config)#int fast
ISLAMABAD(config)#int fastEthernet 0/0
ISLAMABAD(config-if)#ip add 190.1.1.1 255.255.0.0
ISLAMABAD(config-if)#no shutdown
```

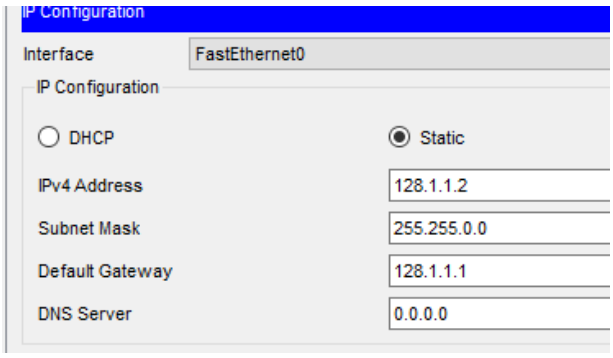
Pc 1 gateway and pc Ip



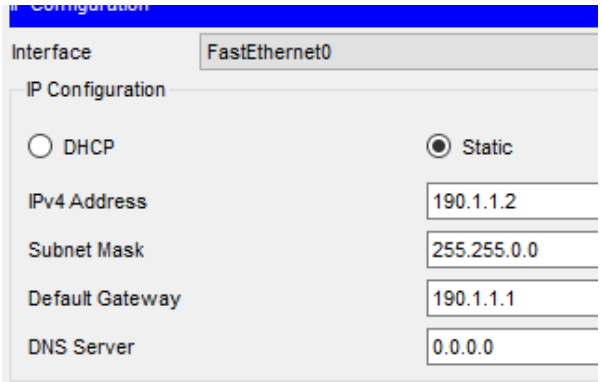
Pc 2 gateway and pc Ip



Pc 3 gateway and pc Ip



Pc 4 gateway and pc Ip



Advertised missing pc Ip in router 1

In R1 pc 2,pc 3,pc 4 Ip is missing

```
Awais(config-if)#ip route 120.1.0.0 255.255.0.0 1.1.1.2
Awais(config)#
```

```
Awais>en
Awais#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Awais(config)#ip route 120.1.0.0 255.255.0.0 130.1.1.2
Awais(config)#ip route 128.1.0.0 255.255.0.0 130.1.1.2
Awais(config)#ip route 190.1.0.0 255.255.0.0 1.1.1.2
Awais(config)#
```

In R2 pc 1, pc 3,pc 4 Ip is missing

```
25(config-if)#ip route 15.0.0.0 255.0.0.0 1.1.1.1
25(config)#
```

```
25>en
25#config t
Enter configuration commands, one per line.  End with CNTL/Z.
25(config)#ip route 190.1.0.0 255.255.0.0 192.168.1.2
25(config)#ip route 128.1.0.0 255.255.0.0 3.1.1.2
25(config)#
```

In R3 pc 4, pc 1,pc 2 Ip is missing

```
F22-BSC-IET-025(config-if)#ip route 190.1.0.0 255.255.0.0 3.1.1.2
F22-BSC-IET-025(config)#ip route 15.0.0.0 255.0.0.0 130.1.1.1
F22-BSC-IET-025(config)#ip route 120.1.0.0 255.255.0.0 3.1.1.2
F22-BSC-IET-025(config)#
```

In R4 pc 2, pc 3, pc 1 Ip is missing

```
ISLAMABAD(config-if)#no shutdown
ISLAMABAD(config-if)#ip route 128.1.0.0 255.255.0.0 3.1.1.1
ISLAMABAD(config)#ip route 120.1.0.0 255.255.0.0 192.168.1.1
ISLAMABAD(config)#ip route 15.0.0.0 255.0.0.0 1.1.1.2
ISLAMABAD(config)#
```

Now pinging pc 1 with pc 2

```
C:\>ping 128.1.1.2

Pinging 128.1.1.2 with 32 bytes of data:

Reply from 128.1.1.2: bytes=32 time=30ms TTL=126
Reply from 128.1.1.2: bytes=32 time=1ms TTL=126
Reply from 128.1.1.2: bytes=32 time=3ms TTL=126
Reply from 128.1.1.2: bytes=32 time=30ms TTL=126
```

Now pinging pc 1 with pc 3

```
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 31ms, Average = 12ms

C:\>ping 128.1.1.2

Pinging 128.1.1.2 with 32 bytes of data:

Reply from 128.1.1.2: bytes=32 time=30ms TTL=126
Reply from 128.1.1.2: bytes=32 time=1ms TTL=126
Reply from 128.1.1.2: bytes=32 time=3ms TTL=126
Reply from 128.1.1.2: bytes=32 time=30ms TTL=126

Ping statistics for 128.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 30ms, Average = 16ms
```

Now pinging pc 1 with pc 4

```
C:\>ping 190.1.1.2

Pinging 190.1.1.2 with 32 bytes of data:

Reply from 190.1.1.2: bytes=32 time=26ms TTL=125
Reply from 190.1.1.2: bytes=32 time=50ms TTL=125
Reply from 190.1.1.2: bytes=32 time=18ms TTL=125
Reply from 190.1.1.2: bytes=32 time=11ms TTL=125

Ping statistics for 190.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 50ms, Average = 26ms
```

Now pinging pc 2 with pc 1

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 15.1.1.2

Pinging 15.1.1.2 with 32 bytes of data:

Reply from 15.1.1.2: bytes=32 time=1ms TTL=126
Reply from 15.1.1.2: bytes=32 time=1ms TTL=126
Reply from 15.1.1.2: bytes=32 time=3ms TTL=126
Reply from 15.1.1.2: bytes=32 time=3ms TTL=126

Ping statistics for 15.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 2ms
```

Now pinging pc 2 with pc 4

```
C:\>ping 190.1.1.2

Pinging 190.1.1.2 with 32 bytes of data:

Reply from 190.1.1.2: bytes=32 time=31ms TTL=126
Reply from 190.1.1.2: bytes=32 time=30ms TTL=126
Reply from 190.1.1.2: bytes=32 time=3ms TTL=126
Reply from 190.1.1.2: bytes=32 time=3ms TTL=126

Ping statistics for 190.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 31ms, Average = 16ms
```

Now pinging pc 2 with pc 3

```
C:\>ping 128.1.1.2

Pinging 128.1.1.2 with 32 bytes of data:

Reply from 128.1.1.2: bytes=32 time=39ms TTL=125
Reply from 128.1.1.2: bytes=32 time=4ms TTL=125
Reply from 128.1.1.2: bytes=32 time=2ms TTL=125
Reply from 128.1.1.2: bytes=32 time=4ms TTL=125

Ping statistics for 128.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 39ms, Average = 12ms

C:\>
```

Now pinging pc 3 with pc 4

```
C:\>ping 190.1.1.2

Pinging 190.1.1.2 with 32 bytes of data:

Reply from 190.1.1.2: bytes=32 time=31ms TTL=126
Reply from 190.1.1.2: bytes=32 time=30ms TTL=126
Reply from 190.1.1.2: bytes=32 time=3ms TTL=126
Reply from 190.1.1.2: bytes=32 time=3ms TTL=126

Ping statistics for 190.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 31ms, Average = 16ms
```

Now pinging pc 3 with pc 1

```
C:\>ping 15.1.1.2

Pinging 15.1.1.2 with 32 bytes of data:

Reply from 15.1.1.2: bytes=32 time=25ms TTL=126
Reply from 15.1.1.2: bytes=32 time=2ms TTL=126
Reply from 15.1.1.2: bytes=32 time=15ms TTL=126
Reply from 15.1.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 15.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 25ms, Average = 10ms
```

Now pinging pc 3 with pc 2

```
C:\>ping 120.1.1.2

Pinging 120.1.1.2 with 32 bytes of data:

Reply from 120.1.1.2: bytes=32 time=28ms TTL=125
Reply from 120.1.1.2: bytes=32 time=35ms TTL=125
Reply from 120.1.1.2: bytes=32 time=2ms TTL=125
Reply from 120.1.1.2: bytes=32 time=2ms TTL=125

Ping statistics for 120.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 35ms, Average = 16ms

C:\>
```

Now pinging pc 4 with pc 2:

```
C:\>ping 120.1.1.2

Pinging 120.1.1.2 with 32 bytes of data:

Reply from 120.1.1.2: bytes=32 time=28ms TTL=126
Reply from 120.1.1.2: bytes=32 time=1ms TTL=126
Reply from 120.1.1.2: bytes=32 time=2ms TTL=126
Reply from 120.1.1.2: bytes=32 time=34ms TTL=126

Ping statistics for 120.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 34ms, Average = 16ms
```

Now pinging pc 4 with pc 3:

PC 4

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 128.1.1.2

Pinging 128.1.1.2 with 32 bytes of data:

Reply from 128.1.1.2: bytes=32 time=1ms TTL=126
Reply from 128.1.1.2: bytes=32 time=2ms TTL=126
Reply from 128.1.1.2: bytes=32 time=2ms TTL=126
Reply from 128.1.1.2: bytes=32 time=3ms TTL=126

Ping statistics for 128.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 2ms
```

Now pinging pc 4 with pc 1:

```
C:\>ping 15.1.1.2

Pinging 15.1.1.2 with 32 bytes of data:

Reply from 15.1.1.2: bytes=32 time=27ms TTL=125
Reply from 15.1.1.2: bytes=32 time=2ms TTL=125
Reply from 15.1.1.2: bytes=32 time=10ms TTL=125
Reply from 15.1.1.2: bytes=32 time=2ms TTL=125

Ping statistics for 15.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 27ms, Average = 10ms

C:\>
```

Conclusion

The lab aimed to create a complicated computer network with four routers and many computers. They wanted to make sure all devices could talk to each other smoothly. In the end they successfully set up the network making sure all the computers and routers could connect and communicate showing their good understanding of how to design and set up networks.